

# Jan Bancerewicz

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## Experience

### ERGO Hestia S.A. – Software Engineering Intern

Jan. 2026 – Currently

- **Develop and maintain** the core insurance premium calculation system - (PricePoint) - that receives 200k+ daily requests.
  - Built and scheduled **data pipelines on Azure Databricks**, integrating corporate data warehouses and ensuring high accuracy for business-critical workflows.
  - Refactored Earnix translation script into a **reusable scikit-learn pipeline**, mapping risk formulas onto **prediction models**.
  - Diagnosed **production incidents** and tracked system health with Dynatrace, tracing failing requests through the pricing API to their root cause.
  - Delivered features through a **CI/CD workflow**, collaborating with the team on code reviews and iterative deployments.
- Tech stack: Python · SQL · Azure · Databricks · API · Git · Agile

### Energia Operator S.A. – IT Department Intern

Jul. 2025 – Dec. 2025

- Administered **OT infrastructure** across a fleet of **90 virtualized servers** in a critical energy distribution environment, provisioning Hyper-V VMs and maintaining system configurations across both **Windows and Linux**.
  - Developed **automation scripts** in PowerShell, Bash and Ansible covering server monitoring, alerting and **backup and recovery** routines, replacing recurring manual checks with scheduled jobs.
  - Containerized internal services with **Docker** and exposed them through an **nginx** reverse proxy, standardising how tools were deployed and restarted.
  - Applied **DevOps practices** day to day: infrastructure as code, version-controlled configuration in Git and reproducible deployments over ad-hoc manual setup.
- Tech stack: Hyper-V · Ansible · PowerShell · Bash · Linux · Windows · Azure · Vim · Git · Nginx

## Education

### Gdańsk University of Technology – B.Sc in Computer Science

2023 – 2027

Final-year student, graduating in February 2027. Overall GPA: 4.41/5.0. Currently working on:

- Engineering Thesis: **Code analysis tool** - a **VS Code extension** performing static analysis of code files to generate extended function flow graphs and code quality metrics based on the parse tree
- Science Club project (Gradient PG): **CORE** - Comments as a REasoning - research project applying mechanistic interpretability to analyze how code comments influence LLM internal activations and reasoning pathways beyond surface-level output quality.

### Electrical Technical School in Białystok – Programmer Technician

2019 – 2023

- **INF.03**: Development and administration of websites, web applications and databases.
- **INF.04**: Designing, programming and testing applications.

Recipient of the Scholarship for Excellence and participant of the Erasmus+ vocational internship program.

## Skills

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|----------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Communication</b>       | Polish – Native, English – Advanced (C1), German – Beginner (A2)                                                         |
| <b>Programming:</b>        | <b>Python, SQL, TypeScript</b> , Java, C++, C#                                                                           |
| <b>ML &amp; Training:</b>  | <b>PyTorch, Hugging Face (Transformers)</b> , scikit-learn, LoRA/PEFT, MLflow, Pandas                                    |
| <b>LLM &amp; RAG:</b>      | vLLM, LlamaIndex, Qdrant, RAG systems, Anthropic/OpenAI API, Google ADK                                                  |
| <b>AI Domains:</b>         | LLM interpretability, Multimodal/VLMs, Computer Vision (YOLO, OpenCV), Signal processing, Agentic AI, Model optimization |
| <b>Backend &amp; Data:</b> | FastAPI, PostgreSQL, Spring Boot, .NET, Redis, MongoDB, REST API, Android                                                |
| <b>Cloud &amp; DevOps:</b> | Docker, Kubernetes, Azure Databricks, AWS, CI/CD (GitHub Actions), Git                                                   |
| <b>Web &amp; GenAI:</b>    | React, Next.js, Tailwind CSS, Claude Code, Cursor IDE                                                                    |

## Certifications

- **Google**, "Umiejętności Jutra AI" program
- **NVIDIA**, Fundamentals of Deep Learning
- **DataCamp**, Python Developer Associate
- **NVIDIA**, Fundamentals of Accelerated Computing with CUDA Python
- **NVIDIA**, Generative AI with Diffusion Models

## Achievements

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- **ICPC Contestant**, [Polish Collegiate Programming Contest \(AMPPZ\)](#) | Representing Gdańsk University of Technology since 2024; placed **#34 nationally** in 2024 and **#55** in 2025 (out of ~80 top university teams). Proved understanding of complex data structures and algorithms in high-pressure problem-solving.
- **#1 place**, [EU Critical Infrastructure Hackathon](#) (2026, Gdańsk) | Won the Energy & Offshore category with an integrity and anomaly-detection layer for PV inverter telemetry. Selected by an international jury of industry experts and investors.
- **#2 place**, [Neurohackathon Heroes of the Brain](#) (2025, Wrocław) | Built a real-time emotion classifier on raw EEG from an 8-channel headband, extracting per-channel frequency and brain-activity patterns, that were used for training ML model.
- **Participant**, [Hackology II](#) (2026, Lublin) | Designed a four-stage computer vision pipeline for retail shelf product recognition: YOLO detection, embedding-based retrieval, heuristic reranking and contextual reranking, evaluated at mAP@0.5.
- **Solo High Score**, [HackYeah](#) (2025, Kraków) | Built a mobile biofeedback app extracting real-time PPG-based heart rate and HRV via the smartphone camera, integrated with an LLM pipeline for actionable stress analysis and personalized reports.

## Activities & Science Clubs

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- **President** (since 2024), [Algorithmic Science Club Sfera PG](#) | Built a community of ambitious students, involved in competitive programming, contests, hackathons and expanding tech opportunities. Mentor peers in technical training (algorithmic problem solving in C++) and organise internal workshops.
- **Board member** (since 2026), [AI/ML Science Club Gradient PG](#) | Contribute to the club's research track and hands-on deep learning projects, working on **ML Autonomous Boat**, deploying real-time YOLO inference on an NVIDIA Jetson edge device.
- **Fundraise project coordinator** (2024), ["Losowanie Świąteczne"](#), ETI Student Faculty Council | Coordinated the annual festive fundraiser at Gdańsk University of Technology for a children's hospice in Gdańsk, collecting over **PLN 9,600**.

## Highlighted Projects

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### LLM Internal States Monitoring: Real-Time Interpretability Instrument [github.com/JanBancerewicz/LLM-monitoring](https://github.com/JanBancerewicz/LLM-monitoring)

Full-stack tool that turns black-box LLMs into observable systems by running local Hugging Face checkpoints in-process and attaching **PyTorch forward hooks** across every decoder layer. Streams **logit lens** projections, KL-to-output convergence, decision depth, attention prompt mass and residual geometry over **SSE** to a **React 19 + FastAPI** dashboard while generation runs, aggregating tensors into scalars inside the hook. Trained per-layer **tuned lens** translators cutting mean KL error against the raw lens by **71.6%** on Qwen3-1.7B and **75.6%** on Gemma 4 E2B-it, and built a cross-checkpoint comparison module indexing each run token × layer in SQLite.

### Coronary Vessel Segmentation with a Multimodal LLM Vision Tower [github.com/JanBancerewicz/arterio-AI-segmentation](https://github.com/JanBancerewicz/arterio-AI-segmentation)

Adapted the **Qwen3-VL 8B** vision tower into a dense segmenter for X-ray coronary angiography, fusing intermediate ViT layers through **DeepStack** and attaching **LoRA/PEFT** adapters ahead of a 6M-parameter CNN decoder. Trained in bf16 on 2×16 GB GPUs and benchmarked on ARCADE SYNTAX against a **ResNet-34 U-Net** baseline under an identical BCE+Dice loss: **Dice 0.758 vs 0.795** at **identical precision** (0.835), with ablations isolating **+0.053** Dice from DeepStack and **+0.086** from LoRA. Evaluated with topology-aware cDice; output masks feed *Arterio*, the coronary angiography labeling platform at Gdańsk Tech.

### Analysis of HRV Using Mobile Devices and Machine Learning [github.com/JanBancerewicz/research-project](https://github.com/JanBancerewicz/research-project)

Research project implementing **PyTorch CNN/LSTM models** for ECG and camera-based PPG signal processing, peak detection and HRV analysis. Models achieved **F1 ≈ 0.98** for ECG R-peak detection and accuracy ≈ 98% for PPG peaks, benchmarked against classical Pan-Tompkins algorithm. The system and results were **published in TASK Quarterly journal** and on GitHub. Link to publication: <https://journal.mostwiedzy.pl/TASKQuarterly/article/view/3699>